

Introduction to Microeconomics

Fall 2025

SEMINAR # 1

3 OCTOBER 2025

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Fundamental principles: opportunity cost

➤ **Opportunity cost:** cost corresponding to the alternative, non-chosen option. A choice implies that you always have to give up something (i.e. an alternative) to obtain a gain. The true cost of something is what you give up in order to get it. This is not an accounting cost.

— EXAMPLE: An entrepreneur decides to set up a business. His forecasts: income CHF 240,000/year, estimated costs (by accountants) 125,000 CHF/year. What would be the profit? Is setting up this business profitable?

— Does the answer change if the person is currently employed at 160,000 CHF/year?

➤ Opportunity cost: the value of the best use of a resource (other than the current use).

Economic profit = Revenues - **Economic costs**

↓
Accounting costs + Opportunity costs

Exercise 1

Which of the following statements is correct? Select one

- a. The opportunity cost is a concept that measures direct monetary costs.
- b. Iris hesitates between self-driving and public transportation for a holiday travel. The opportunity cost of self-driving is equal to the cost of public transportation.
- c. The opportunity cost accounts for alternative uses of resources such as time and money.
- d. The opportunity cost of buying a property is always smaller than the local rental rate for a comparable house.

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Exercise 2

Which of the following statements is correct? Select one

- a. The opportunity cost of leisure includes the wage that could be earned by working instead.
- b. The opportunity cost of attending university includes the future pension that the individual will obtain upon retiring.
- c. The opportunity cost of buying a new car is always smaller than the local rental rate for a comparable car.
- d. The opportunity cost of pursuing a university degree is higher in recessions (when the unemployment rate rises) than in economic booms.

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Microeconomics *versus* Macroeconomics (1)

MICROECONOMICS

- studies the decision process of the representative/typical economic agent (producers, consumers, savers, workers, investors, workers... and government)
- focuses on behavior and interactions between agents (exchange on a market, contracts...)

MACROECONOMICS

- considers the economy as a whole and focuses on “aggregates”, such as production [GDP], consumption, growth, inflation, unemployment
- analyses how the level of these aggregates is determined and how they influence each other

Exercise 3

Decide whether the following topics are part of micro- or macroeconomics:

1. A family choosing how much to save.
2. Impact of public regulation on car pollution.
3. Effect of an increase of the savings rate on economic growth.
4. A company choosing the number of its employees.
5. The relationship between the inflation rate and the variation in money supply.

Exercise 3

Decide whether the following topics are part of micro- or macroeconomics:

1. A family choosing how much to save. **Micro**
2. Impact of public regulation on car pollution. **Micro**
3. Effect of an increase of the savings rate on economic growth. **Macro**
4. A company choosing the number of its employees. **Micro**
5. The relationship between the inflation rate and the variation in money supply. **Macro**

Correlation *versus* causality (1)

CORRELATION: two events vary together

CAUSALITY: one event directly impact another event

PROBLEM: variables can vary together for whatever reason, or even for no reason!

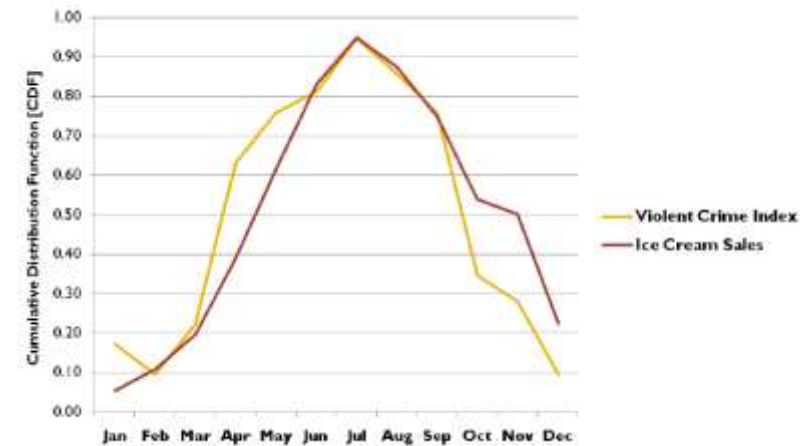
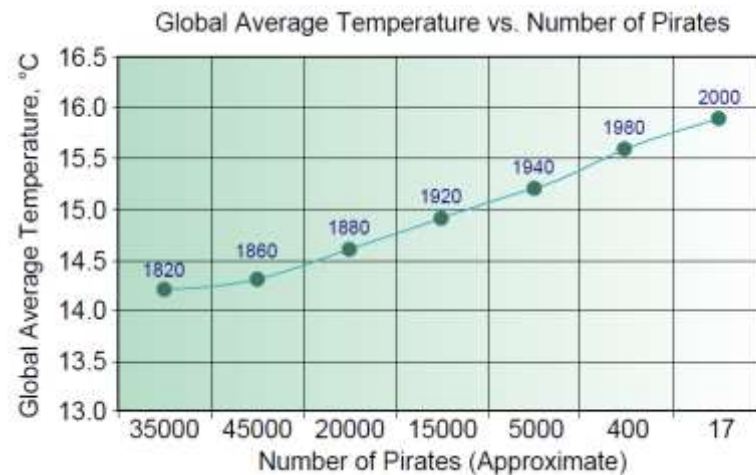
PINGO: Go to <https://pingo.coactum.de/> and digit the code 294526

Among the following propositions, which ones are correct? [several possible answers]

- A. Most people die in their bed, which is evidence that the horizontal position is lethal.
- B. Named cows produce more milk, which is evidence that giving names to cows improves their health and productivity.
- C. Gamblers over 65 years of age are healthier than non-gamblers of the same age group, which is evidence that gambling is healthy.
- D. High poverty rates of countries under IMF adjustment programs is evidence that IMF programs impoverish countries.
- E. Relatively high crime rates of people of foreign origin is evidence that foreigners are more criminal than natives.

Correlation *versus* causality (2)

EXAMPLE (correlation): "The mortality rate is higher in a hospital bed than in your own bed." (positive correlation between mortality and night spent in the hospital).



And many others... You can create many other examples of **spurious correlation** on your own [here](#).

PROBLEM: Misattribution of causality can lead to very bad **public policies** (e.g. eliminating pirates to solve global warming or banning the sale of ice cream??).

Exercise 4

On average, pupils do better at school (have better test results) if they study in smaller classes with fewer classmates and are taught by teachers that are paid more. Which of the following conclusions is correct?

- a. There exists a positive correlation between the salaries of teachers and the test results of the pupils as well as a negative correlation between class size and the test results of the pupils.
- b. The observation proves that the salaries of teachers and class size influence student performance.
- c. The observation proves that the best teachers, who are also paid the most, prefer working in smaller classes.
- d. The observation proves nothing.

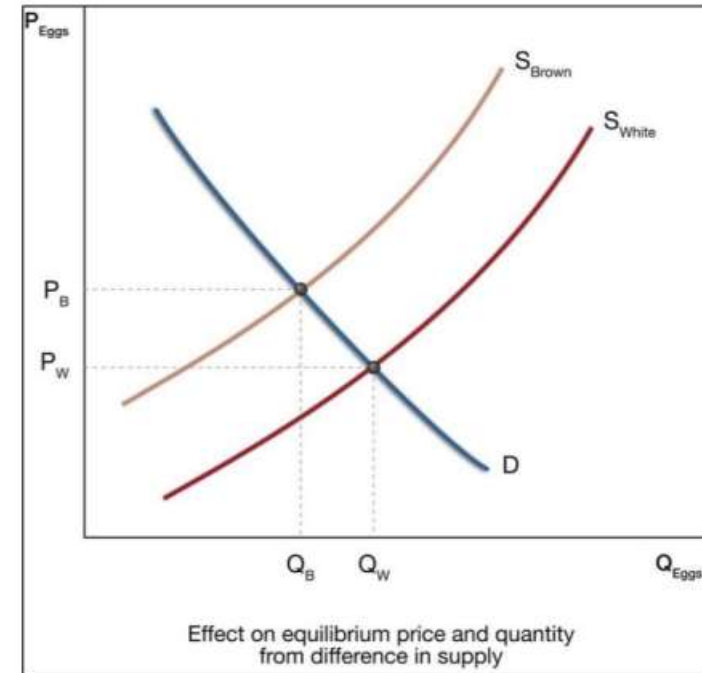
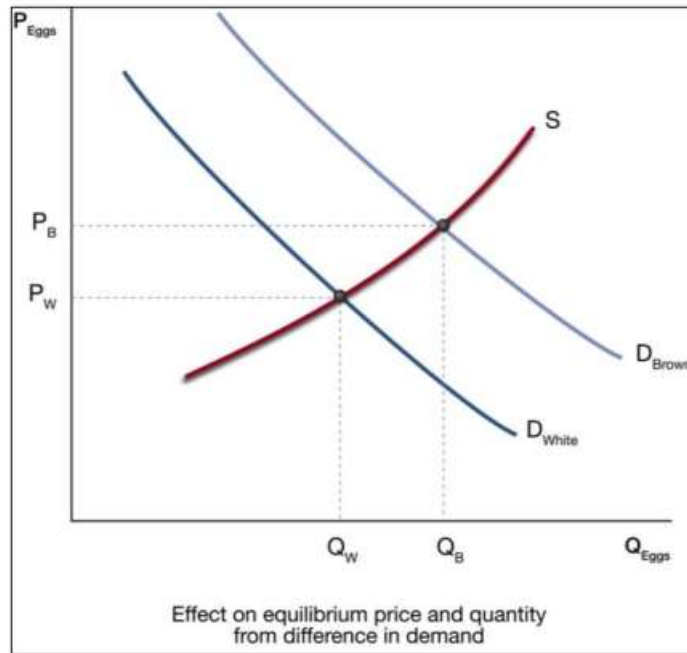
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Exercise 5

Why do brown eggs cost more than conventional ones (white eggs)?



Exercise 6:

Using a supply-and-demand diagram, answer the following questions. First, why do the prices of roses increase right before Valentine's Day?

- A. This outcome is the result of a move of the equilibrium price and quantity along the supply curve.
- B. This outcome is the result of an outward shift of the demand curve.
- C. This outcome is the result of a shift of the supply curve.
- D. This outcome is the result of a shift of both demand and supply.

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Exercise 7:

Consider the following supply and demand functions:

$$QS(P) = P - 1 \text{ and } QD(P) = 5 - P$$

Where QS is the amount of a good supplied on the market, QD is the amount of a good demanded on the market and P is the price of this good in this market.

Assume that due to an increase in production costs the supply function has changed as follows: $QS(P) = P - 3$

It follows that after the increase in production costs:

- a. the equilibrium price has increased from 1 to 3.
- b. producers' revenue has increased from 3 to 4.
- c. the quantity traded on the market has decreased from 2 to 1.
- d. total spending on this good has increased from 4 to 6

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Substitutes and complements (3)

- **SUBSTITUTES** (definition): two goods are **substitutes** if an increase in the price of one good leads to an increase in the quantity demanded of the other good
- **COMPLEMENTS** (definition): two goods are **complements** if an increase in the price of one good leads to a decrease in the quantity demanded of the other good.
- Imagine a demand curve shifts to the right:
 - A shift from D to D' could be explained by an **increase** in the price of a substitute good.
 - A shift from D to D' could be explained by a **decrease** in the price of a complement good.

Exercise 8:

The market for wheat is initially in equilibrium. What are the overall consequences of a negative shock to supply caused by heavy rains and of an increase of the price of corn (which is considered to be a substitute for wheat)?

- a. The price of wheat increases but the effect on the quantity exchanged is undetermined.
- b. The price of wheat and the quantity exchanged increase.
- c. The price of wheat increases and the quantity exchanged decreases.
- d. Nothing can be said about the direction of the changes in the quantity traded and in price.

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If time permits...

Exercise 9: You are on considering the following alternatives. The cost of going to a concert, A, in a theatre, has a \$25 admission cost. In a nearby park there is concert B, which is free but happens at the same time. Suppose that the most you would have been willing to pay to attend the free concert in the park (if it was not free) was \$15. Which of the following statements are correct?

- A. The benefit of your next best alternative to concert A would be \$15 of enjoyment in the park. It is then the opportunity cost of going to concert A.
- B. The total economic cost of concert A is $\$25 + \$15 = \$40$.
- C. If you anticipate \$35 worth of pleasure from concert A, then you will choose not to go to the theatre.
- D. If you anticipate \$50 worth of pleasure from concert A, then you will choose to go to the theatre.

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All are correct

Exercise 10:

An economic researcher collects data on the wage and productivity of grape pickers. She finds that workers who collect many grapes per hour tend to be better paid than those who collect only a few, *ceteris paribus*. What can you safely conclude from observing this fact?

- A. Highly paid employees are more productive because they are better satisfied with their salary.
- B. Employers reward productive employees with higher salaries.
- C. The salary of an employee and her productivity are positively correlated.
- D. The number of grapes picked per hour is a misleading measure of productivity

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Exercise 11:

You are a taxi driver in Melbourne who earns A\$50 for a day's work. You have been offered a one-day ticket to the Australian Open for A\$40. As a tennis fan, you value the experience at A\$100. With this information, what can we say?

- A. The opportunity cost of the day at the Open is A\$40.
- B. The economic cost of the day at the Open is A\$40.
- C. The economic rent of the day at the Open is A\$10.
- D. You would have paid up to A\$100 for the ticket

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Exercise 12:

It is observed that married men live longer than single men. Can we safely conclude that marriage increases life expectancy?

- A. Yes, this relationship is causal.
- B. No, the causality goes in reverse: a longer life expectancy increases the probability that a man will wed in his lifetime.
- C. No, because there is an omitted variable bias: healthier people might at once enjoy a longer life expectancy and a greater likelihood of getting married.
- D. No, because the probability of getting married varies with age and this leads to mean reversion

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The so-called market price is the price at which the quantity supplied is equal to the quantity demanded. Sometimes, prices do not exist and/or are different from the market clearing price. Excess supply or excess demand ensues.

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- B. Waiting lists for rental apartments.
- C. Black market for concert tickets.
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- E. Minimum wage-led unemployment

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